ATOM COUNT CARD

EXPLORING THE MOLE CONCEPT THROUGH VINEGAR

A. PRODUCT IDENTITY



Product name: Silver Swan Sukang Puti

Product type: Naturally Fermented

Compound name: Acetic Acid

Molecular formula: $C_2H_4O_2$

B. BASE QUANTITATIVE DATA

Quantity	Value	Source	Precision
Net content	200 mL	Product label	Nominal
Formula $C_2H_4O_2$	subscripts 2, 4, 2	NIST	Exact count
Density (Assumption)	1.01 g/mL	Chemistry Textbook	3 significant figure
Acidity (Assumption)	4.5% (w/v)	Landers	Nominal
Serving Size	1 tbsp (15mL)	Product Label	Nominal

C. MOLAR MASS BUILD-UP

Elements	Atoms	Atomic mass	g/mol	% percent by mass
C	2	12.01	24.022	40.00%
H	4	1.008	4.032	6.71%
O	2	15.999	31.998	53.29%
$C_2H_4O_2$	8	-	60.052	100.00%

D. MOLE MAP

200 mL (× 4.5g / 100mL) → Mass: 9.0g (×1 mol / 60.052g) → Moles: 0.15 mol (× 6.02×10^{23} molecules / 1 mol) → 9.03×10^{22} molecules (× 2 C atoms / 1 molecules) → 1.806×10^{23} C atoms (× 1 mol C / 6.02×10^{23} × (12.011 g C / 1 mol C) ≈ 3.60 g C) → 3.60 g C

Assumption: The 4.5% acidity is based on an external reference, equivalent to 4.5 g of acetic acid per 100 mL of vinegar (w/v).

Mass of Carbon ← Carbon Atoms ← Particles

E. INSIDE THE FORMULA

1. Molar Mass of $C_2H_4O_2$

$2(12.011) + 4(1.008) + 2(15.999) = 60.052$ g/mol

2. Label Bridge: amount of compound in the bottle

$$200\ \cancel{mL}\ \text{vinegar} \times \frac{4.5g}{100\cancel{mL}} = 9.0g$$

3. Mass to moles

$$9.0\cancel{g} \times \frac{1mol}{60.052\cancel{g}} = 0.14987\ \text{mol} \rightarrow \textbf{0.15 mol}$$

4. Moles to molecules

$$0.14987\ \cancel{\text{mol}} \times \frac{6.02 \times 10^{23}\ \text{molecules}}{1\ \cancel{\text{mol}}} = 9.026 \times 10^{22}\ \text{molecules} \rightarrow \textbf{9.03 × 10²² molecules}$$

5. Number of atoms of one element

$$9.026 \times 10^{22}\ \cancel{\text{molecules}} \times \frac{2\ \text{C atoms}}{1\ \cancel{\text{molecule}}} = 1.8052 \times 10^{23}\ \text{C atoms} \rightarrow \textbf{1.8 × 10²³ C atoms}$$

6. Percent Composition and two-route check

$$\%C = \frac{24.022\ \cancel{g/mol}}{60.052\ \cancel{g/mol}} \times 100 = 40.00\%$$

Route A:

$$9.0g\ \cancel{\text{acetic acid}} \times \frac{40.00g\ C}{100\ g\ \cancel{\text{acetic acid}}} = 3.6g\ C$$

Route B:

$$0.14987\ \cancel{\text{mol acetic acid}} \times \frac{2\ \cancel{\text{mol C}}}{1\ \cancel{\text{mol acetic acid}}} \times \frac{12.011g\ C}{1\ \cancel{\text{mol C}}} = 3.600g\ C$$

Both Routes → 3.6g C ✓ **Agree**

7. Reverse: From % composition → Formula

In 100 g acetic acid (assumption)

C 40.00 ÷ 12.011 = 3.3303 mol; H 6.71 ÷ 1.008 = 6.6567 mol; O 53.29 ÷ 15.999 = 3.3308 mol

÷ 3.3303 → 1 : 2 : 1 → **CH_2O** (empirical = molecular, n = 2)

8. Reverse: How much solution is one mole?

$$1.00\ \cancel{\text{mol}} \times \frac{60.052\ \cancel{g}}{1\ \cancel{\text{mol}}} \times \frac{100\ \text{mL solution}}{4.5\ \cancel{g}}$$

= 1334.49 mL solution → **1.3×10^3 mL solution**

F. SCALE UP

1. **One Pouch**
   - 200 mL vinegar solution
   - 9.0 g acetic acid
   - 0.1499 mol acetic acid
   - 9.026×10^{22} molecules

2. **One Laboratory Session**
   Assume 36 students × 1 pouch / student
   Total Acetic acid: 36 × 9.0g = 324g
   Total Moles: 36 × 0.14987 = 5.4 mol
   Total Pouches: 36 × 10 = 360 pouches

3. **One Academic Year**
   Assume 36 students × 1 pouch each × 10 laboratory session
   Total Molecules: $36 \times 9.026 \times 10^{22} = 3.2 \times 10^{24}$
   Total Volume: 360 × 200 = 72,000 mL
   Total Acetic Acid: 360 × 9.0 = 3240g = 3.24kg
   Total Moles: 360 × 0.14987 ≈ 54 mol

4. **Reverse: How many pouches make one mole?**

$$\frac{1.00\ \cancel{\text{mol}}}{0.14987\ \cancel{\text{mol}}\ \text{/ pouch}} = 6.67\ \text{pouches}$$

G. SIGNIFICANT FIGURES CHECK

Quantity	Calculated Values	Reported Values
Moles per pouch	0.14987 mol	0.15 mol
Molecules per pouch	9.026×10^{22}	9.0×10^{22}
Volume for one pouch	1334.49 mL	1.3×10^{22}

The reported values use 2 significant figures becuase the assumed acidity, 4.5%, has 2 significant figures.

ONE-NUMBER TAKEAWAY

One 200 mL pouch of vinegar contains approximately 9.0×10^{22} acetic acid molecules, assuming 4.5g of acetic acid per 100 mL of solution.

H. REFERENCES

- AAA Shopping Center. (n.d.). Silver Swan Sukang Puti DOY 200ml. https://aaashoppingcenter.ph/products/silver-swan-sukang-puti
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